

Software Test Plan (STP)

Online Internship & Placement Preparation Portal

Field	Detail
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Team	Team 12
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Revision History

Version	Date	Author(s)	Change Summary
1.0	16-09-2026	Team 12	Initial test plan covering all SRS functional, non-functional and security requirements.

1. Introduction

Purpose: This document defines the test plan for the Online Internship & Placement Preparation Portal v1.0. It states the testing objectives, scope, strategy, environment, schedule, deliverables and responsibilities, and establishes traceability between the requirements in the SRS v1.0 and the test cases that verify them.

Scope: Testing covers student and recruiter registration and authentication, internship listing creation and search, application submission and status tracking, the timed aptitude test module, student progress dashboards, admin placement analytics, and the admin/recruiter management panel. It also covers the non-functional requirements IPP-NF-001 to IPP-NF-005 and the security requirements IPP-SR-001 to IPP-SR-005. Third-party email delivery infrastructure, MongoDB Atlas internals and browser internals are outside the scope of this plan.

References: SRS – Online Internship & Placement Preparation Portal v1.0; IEEE 829 Test Documentation Standard; OWASP Top 10 (2021); WCAG 2.1 Level AA.

Definitions: SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), FR (Functional Requirement), NFR (Non-Functional Requirement), JWT (JSON Web Token), OTP (One-Time Password), TLS (Transport Layer Security), RBAC (Role-Based Access Control), UAT (User Acceptance Testing), SLA (Service Level Agreement), PII (Personally Identifiable Information).

2. Test Items

The following build components of the portal (MERN stack) are the items under test:

- Authentication module (registration, email verification, login, OTP password reset, account logout)
- Listing module (internship posting, search, filtering, listing details, preference notifications)
- Application module (application submission, status tracking, recruiter review, notifications)
- Test module (timed aptitude tests, auto-grading, attempt history)
- Analytics module (student dashboard, skill-gap indicator, admin placement analytics, recommendations)
- Admin module (recruiter approval/rejection, listing and user account administration)

- REST API layer (Node/Express endpoints over HTTPS, JWT session handling, RBAC enforcement)
- Python scoring/analytics scripts invoked by the backend
- React web UI across desktop and mobile browsers

3. Features to be Tested

Features are listed against their SRS requirement IDs.

3.1 Functional Features

Req ID	Feature	Priority	Test Case(s)
IPP-F-001	Student registration with SRN, college email and email verification	High	TC-Auth-01
IPP-F-002	Recruiter registration held in pending state until admin verification	High	TC-Auth-02
IPP-F-003	JWT-based secure login	High	TC-Auth-03
IPP-F-004	Forgotten-password reset via single-use OTP	Medium	TC-Auth-04
IPP-F-005	Account lockout for 15 minutes after 5 failed login attempts	High	TC-Auth-05
IPP-F-006	Verified recruiter posts internship listing	High	TC-List-01
IPP-F-007	Search and filter internships by skill, location, stipend, duration	High	TC-List-02
IPP-F-008	Full internship details with eligibility and deadline	Medium	TC-List-03
IPP-F-009	Notification of new internships matching saved skill preferences	Medium	TC-List-04
IPP-F-010	Apply to internship with resume and optional cover letter	High	TC-App-01
IPP-F-011	Real-time application status tracking	High	TC-App-02
IPP-F-012	Recruiter reviews applications and updates status	High	TC-App-03
IPP-F-013	Notification on application status change	Medium	TC-App-04
IPP-F-014	Timed aptitude tests by topic with auto-submit	High	TC-Test-01
IPP-F-015	Auto-grading and immediate score display	High	TC-Test-02
IPP-F-016	Test attempt history with score and time taken	Medium	TC-Test-03
IPP-F-017	Student dashboard with applications, scores and skill-gap indicator	Medium	TC-Ana-01
IPP-F-018	Admin placement analytics report	Medium	TC-Ana-02
IPP-F-019	Internship recommendations based on test performance and skills	Low	TC-Ana-03
IPP-F-020	Admin approves or rejects pending recruiter registrations	High	TC-Admin-01
IPP-F-021	Admin adds, edits or deactivates listings and user accounts	Medium	TC-Admin-02

3.2 Non-Functional and Security Features

Req ID	Feature	Priority	Test Case(s)
IPP-NF-001	Search and page load $\leq 3s$ for 90% of requests under normal load	High	TC-Perf-01
IPP-NF-002	99.5% monthly availability excluding scheduled maintenance	High	TC-Ops-01
IPP-NF-003	Passwords stored using salted bcrypt hashing	High	TC-Sec-01
IPP-NF-004	Support for 5,000 concurrent users without degradation	Medium	TC-Perf-02
IPP-NF-005	Responsive UI and WCAG 2.1 AA basics	Medium	TC-UX-01
IPP-SR-001	TLS 1.2+ mandatory on all client-server communication	High	TC-Sec-02
IPP-SR-002	Passwords hashed with bcrypt and salt, never in plaintext	High	TC-Sec-01
IPP-SR-003	RBAC enforced on every API endpoint	High	TC-Sec-03
IPP-SR-004	Test content served via secure, time-bound tokens	High	TC-Sec-04

Req ID	Feature	Priority	Test Case(s)
IPP-SR-005	Authentication and admin actions logged and retained for 1 year	Medium	TC-Sec-05

4. Features Not to be Tested

Item Excluded	Justification
Internal delivery infrastructure of the third-party email service (SendGrid/SMTP)	Vendor responsibility; only the portal's request/retry behaviour and delivery callback are tested.
MongoDB Atlas engine internals, replication and backup mechanisms	Managed cloud service assumed reliable; only application-level data operations are tested.
Browser rendering engine defects	Out of team control; testing is limited to the supported browser matrix listed in Section 6.
Payroll or stipend disbursement	Explicitly excluded from product scope in SRS Section 1.2.
Official placement-cell record systems	The portal does not replace or integrate with them in this release (SRS Section 1.2).
Node.js, Express and React framework internals	Third-party libraries assumed tested upstream.
Penetration testing of the hosting cloud provider infrastructure	Provider responsibility; only application-layer security is validated.

5. Test Approach / Strategy

5.1 Test Levels

Level	Description
Unit testing	Jest tests on individual Express controllers, React components, Mongoose models and the Python scoring functions. Target $\geq 70\%$ statement coverage on business-logic modules.
Integration testing	Supertest on REST endpoints against a test MongoDB instance; backend $\leftrightarrow$ Python scoring script; backend $\leftrightarrow$ email service (stubbed); React $\leftrightarrow$ API contract checks.
System testing	End-to-end journeys through the browser: register $\rightarrow$ verify $\rightarrow$ search $\rightarrow$ apply $\rightarrow$ track; recruiter post $\rightarrow$ review $\rightarrow$ status update; admin verify $\rightarrow$ analytics.
Acceptance testing (UAT)	Executed by placement coordinators and a sample student/recruiter group against the acceptance criteria AC-001 to AC-021 in the SRS.
Regression testing	Automated smoke and regression suite re-executed on every build after defect fixes.

5.2 Test Types

- Functional testing — verification of all FRs IPP-F-001 to IPP-F-021 against their acceptance criteria.
- Negative and boundary testing — invalid SRN/email formats, expired OTP, duplicate applications, timer boundary at 00:00, filter combinations returning empty sets.
- Performance and load testing — JMeter runs measuring 90th-percentile response time (IPP-NF-001) and a 5,000 virtual-user soak (IPP-NF-004).
- Security testing — per Section 5.1 (Security Validation) below.
- Usability and accessibility testing — WCAG 2.1 AA checks using axe DevTools and manual keyboard/screen-reader walkthroughs.
- Compatibility testing — Chrome, Firefox, Edge and Safari on desktop, plus Android Chrome and iOS Safari on mobile.
- Regression testing — Selenium suite covering all High-priority requirements.

5.3 Entry Criteria

- SRS v1.0 reviewed and baselined.

- Stable build deployed to the test environment and passing the smoke suite.
- Test cases written, reviewed and approved; test data (student accounts, recruiter accounts, sample listings, question bank) loaded.
- Test environment, tools and defect-tracking project configured and accessible.

5.4 Exit Criteria

- 100% of planned test cases executed.
- 0 critical and 0 high-severity defects open; medium/low defects triaged with an agreed disposition.
- All High-priority functional requirements passed, and no critical NFR failure (IPP-NF-001 and IPP-NF-003 must pass).
- RTM shows every requirement mapped to at least one executed test case with a recorded result.
- Test Summary Report prepared and signed off by the QA Lead and Product Owner.

5.5 Security Validation

Security validation addresses the security objectives in SRS Section 5.1.1 and verifies requirements IPP-SR-001 to IPP-SR-005 and IPP-NF-003.

Activity	Description	Requirement
Password storage audit	Inspect the users collection directly after registration and password reset to confirm only salted bcrypt hashes are persisted, with no plaintext or reversible encoding in the database, logs or API responses.	IPP-NF-003, IPP-SR-002
Transport security check	Run an SSL/TLS scan (SSL Labs / testssl.sh) on the deployed endpoint to confirm TLS 1.2+ is enforced, plaintext HTTP redirects to HTTPS, and weak ciphers/protocols are disabled.	IPP-SR-001
Role-based access control testing	Attempt every protected API endpoint with no token, an expired token, a tampered JWT signature, and a token of the wrong role (student → admin endpoints, recruiter → other recruiters' applications). All must return 401/403.	IPP-SR-003
Test-content token validation	Verify aptitude test content is served only via time-bound tokens; attempt reuse of a consumed token and use after expiry; confirm answers are not present in the client payload before submission.	IPP-SR-004
Audit log verification	Confirm login, failed login, password reset, recruiter approval/rejection and listing deactivation events are written to the audit log with actor, action and timestamp, and that retention is configured for ≥ 1 year.	IPP-SR-005
Brute-force / lockout testing	Issue repeated failed logins to confirm the 15-minute lockout after 5 attempts and verify the lockout is enforced server-side, not only in the UI.	IPP-F-005
Input fuzzing and injection testing	Fuzz registration, login, search filter and listing forms with NoSQL injection payloads, XSS strings and oversized inputs; verify parameterised queries, input sanitisation and output encoding.	OWASP A03
File upload validation	Upload oversized files, executables and files with spoofed MIME types to the resume field; confirm type/size validation and that uploads are not served from a web-executable path.	IPP-F-010, OWASP A04
Session management review	Verify JWT expiry, that tokens are invalidated on password reset, and that OTPs are single-use and expire in 10 minutes.	IPP-F-004, IPP-SR-003
PII exposure review	Confirm resumes and contact details are accessible only to the owning student, the recruiter of the applied listing, and	SRS 5.1.1

Activity	Description	Requirement
	admins.	

6. Test Environment

Component	Specification
Client hardware	Standard desktop/laptop (8 GB RAM) and Android/iOS mobile devices for responsive testing.
Browsers	Chrome (latest), Firefox (latest), Edge (latest), Safari 16+, Android Chrome, iOS Safari.
Server	Node.js 18 LTS with Express, deployed on a cloud VM; separate staging instance isolated from production.
Database	MongoDB Atlas test cluster seeded with anonymised test data.
Auxiliary services	Python 3.x scoring/analytics scripts; email service sandbox (SendGrid test mode / Mailtrap) for OTP and notifications.
Tools	Selenium WebDriver (UI automation), Jest + Supertest (unit/API), Postman (manual API), JMeter (performance/load), axe DevTools (accessibility), testssl.sh + OWASP ZAP (security), Jira (defect tracking), GitHub Actions (CI).
Test data	50 dummy student accounts, 10 recruiter accounts (5 verified, 5 pending), 1 admin account, 100 sample internship listings, an aptitude question bank of 200 questions, and sample resume files in PDF/DOCX.

7. Test Schedule

Milestone	Start	End
Test plan preparation and review	16-Sep-2026	18-Sep-2026
Test case design and review	18-Sep-2026	20-Sep-2026
Test environment and data setup	20-Sep-2026	21-Sep-2026
Unit and integration test execution	21-Sep-2026	23-Sep-2026
System test execution	23-Sep-2026	25-Sep-2026
Defect fixing and regression	25-Sep-2026	26-Sep-2026
Performance and security testing	26-Sep-2026	27-Sep-2026
User Acceptance Testing (UAT)	27-Sep-2026	28-Sep-2026
Test summary report and sign-off	28-Sep-2026	28-Sep-2026

8. Test Deliverables

- Test Plan (this document)
- Test case specification (manual and automated), including the cases in Appendix A
- Automation scripts (Selenium, Jest/Supertest, JMeter test plans)
- Test data sets and seeding scripts
- Test execution logs and daily status reports
- Defect reports logged in Jira with severity and priority
- Requirements Traceability Matrix (Section 13)
- Performance test results and security validation report
- Test Summary Report with sign-off

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Benita Wilson	Prepare and maintain the test plan, review test cases, coordinate execution, report status, own exit-criteria sign-off.
Test Engineer (Functional)	Bhakti S Patil	Design and execute functional and regression test cases for auth, listings and applications; log and retest defects.
Test Engineer (NFR & Security)	Bharath K	Design and execute performance, security and accessibility tests; maintain JMeter and ZAP configurations.
Developer / Build Owner	Bhavani Shahapur	Deliver stable builds, maintain the test environment, triage and fix defects, support root-cause analysis.
Product Owner / Placement Coordinator	Bhavani Shahapur	Approve acceptance criteria, participate in UAT, sign off release readiness.

10. Risks and Mitigation

Risk	Impact	Mitigation
Delay in delivery of a stable build	High	Request early smoke builds; run CI on every merge; prioritise High-priority test cases first.
Email service sandbox rate limits block OTP and notification testing	Medium	Use Mailtrap as a backup sink and stub the email adapter for high-volume runs.
Test environment downtime or Atlas quota exhaustion	Medium	Maintain a local Docker MongoDB fallback and a documented re-seed script.
5,000-user load test exceeds academic-tier cloud capacity	Medium	Run a scaled load test (e.g. 1,000 users) and extrapolate, documenting the deviation in the summary report.
Incomplete or unrealistic test data (resumes, question bank)	Medium	Prepare and review the data set during environment setup, before execution starts.
Limited team availability due to concurrent academic deadlines	High	Front-load automation, allocate fixed daily testing slots, and keep a shared execution tracker.
Requirement changes after baseline	Medium	Any change routed through revision control on the SRS with an RTM update before test cases are altered.

11. Assumptions & Dependencies

- The SRS v1.0 is baselined; any change is formally versioned and reflected in the RTM.
- The staging environment mirrors the production configuration (Node version, TLS settings, database schema).
- The email service sandbox is available for the full execution window.
- Anonymised test data, including sample resumes and the aptitude question bank, is supplied before execution begins.
- Developers are available for defect triage within one working day during the execution window.
- The Python scoring scripts expose a stable interface to the Node backend.
- Browser versions do not change materially during the test window.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 continuous hours.
- The build is too unstable — a defect blocks more than 30% of planned test cases.
- A critical defect in authentication or data security is found that invalidates dependent test cases.
- Test data is corrupted or lost and cannot be re-seeded.

Resume testing when:

- The blocking defects are fixed, verified and a fresh build is deployed.
- The environment is restored and the smoke suite passes.
- Test data has been re-seeded and validated.
- The QA Lead formally records the resumption in the daily status report.

13. Test Case Management & Traceability

Test cases are maintained in Jira (Xray/Zephyr) and version-controlled alongside the automation code. Every SRS requirement maps to at least one test case; every test case references exactly one primary requirement ID. The RTM below is the authoritative mapping and is updated after each execution cycle. Status legend: N = Not executed, P = Passed, F = Failed.

Req ID	Requirement (short)	SRS Ref	Module	Test Case(s)	Status
IPP-F-001	Student registration & email verification	4.1	AuthModule	TC-Auth-01	N
IPP-F-002	Recruiter pending until admin verification	4.1	AuthModule	TC-Auth-02	N
IPP-F-003	Secure login (JWT)	4.1	AuthModule	TC-Auth-03	N
IPP-F-004	Password reset via OTP	4.1	AuthModule	TC-Auth-04	N
IPP-F-005	Account lockout after 5 failed attempts	4.1	AuthModule	TC-Auth-05	N
IPP-F-006	Post internship listing	4.2	ListingModule	TC-List-01	N
IPP-F-007	Search & filter internships	4.2	ListingModule	TC-List-02	N
IPP-F-008	View full internship details	4.2	ListingModule	TC-List-03	N
IPP-F-009	Skill-preference match notification	4.2	ListingModule	TC-List-04	N
IPP-F-010	Submit application with resume	4.3	ApplicationModule	TC-App-01	N
IPP-F-011	Track application status	4.3	ApplicationModule	TC-App-02	N
IPP-F-012	Recruiter updates application status	4.3	ApplicationModule	TC-App-03	N
IPP-F-013	Status-change notification	4.3	ApplicationModule	TC-App-04	N
IPP-F-014	Timed aptitude test with auto-submit	4.4	TestModule	TC-Test-01	N
IPP-F-015	Auto-grade and display score	4.4	TestModule	TC-Test-02	N
IPP-F-016	Test attempt history	4.4	TestModule	TC-Test-03	N
IPP-F-017	Student dashboard & skill gap	4.5	AnalyticsModule	TC-Ana-01	N
IPP-F-018	Placement analytics report	4.5	AnalyticsModule	TC-Ana-02	N
IPP-F-019	Internship recommendations	4.5	AnalyticsModule	TC-Ana-03	N
IPP-F-020	Approve/reject recruiter accounts	4.6	AdminModule	TC-Admin-01	N
IPP-F-021	Manage listings & user accounts	4.6	AdminModule	TC-Admin-02	N
IPP-NF-001	Response time ≤ 3s (90th percentile)	5	WebUI / API	TC-Perf-01	N
IPP-NF-002	99.5% monthly availability	5	Infrastructure	TC-Ops-01	N

Req ID	Requirement (short)	SRS Ref	Module	Test Case(s)	Status
IPP-NF-003	Salted bcrypt password storage	5	AuthModule	TC-Sec-01	N
IPP-NF-004	5,000 concurrent users	5	API / DB	TC-Perf-02	N
IPP-NF-005	Responsive UI + WCAG 2.1 AA	5	WebUI	TC-UX-01	N
IPP-SR-001	TLS 1.2+ enforced	5.1.2	API / Transport	TC-Sec-02	N
IPP-SR-002	Password hashing (bcrypt + salt)	5.1.2	AuthModule	TC-Sec-01	N
IPP-SR-003	RBAC on every endpoint	5.1.2	API Layer	TC-Sec-03	N
IPP-SR-004	Time-bound test-content tokens	5.1.2	TestModule	TC-Sec-04	N
IPP-SR-005	Audit logging retained 1 year	5.1.2	AuditModule	TC-Sec-05	N

14. Test Metrics & Reporting

Metrics collected:

- Test case execution progress (% planned cases executed)
- Pass/fail rate (% passed, % failed, % blocked)
- Requirement coverage (% of SRS requirements with at least one executed test case)
- Defect density (defects per module / per requirement)
- Defect severity distribution (critical, high, medium, low)
- Defect aging and reopen rate
- Automation coverage (% of regression cases automated)
- Performance metrics (90th-percentile response time, throughput, error rate under load)

Reports:

- Daily test execution status report during the execution window
- Weekly defect trend and RTM coverage snapshot
- Performance test report and security validation report
- Final Test Summary Report with exit-criteria assessment and release recommendation

15. Approvals

Role	Name	Signature / Date
QA Lead	Benita Wilson	
Test Engineer (Functional)	Bhakti S Patil	
Test Engineer (NFR & Security)	Bharath K	
Dev Lead	Bhavani Shahapur	
Product Owner	Bhavani Shahapur	

Appendix A: Detailed Test Cases

The following test cases cover both functional and non-functional requirements. Each case is traceable to a requirement ID in the SRS.

TC ID	Req ID	Objective	Preconditions	Test Steps	Expected Result	Type / Priority
TC-Auth-01	IPP-F-001	Student registration with email verification	Student is not registered; college email inbox accessible; SMTP sandbox running.	1. Open the registration page. 2. Enter SRN PES2UG24CS999, a valid college email, and a valid password. 3. Submit the form. 4. Check the inbox for the verification email. 5. Attempt login before clicking the link. 6. Click the verification link and log in again.	Verification email received within 2 minutes. Login before verification is rejected with "Account not verified". After clicking the link the account is activated and login succeeds. (AC-001)	Functional / High
TC-Auth-03	IPP-F-003	Secure login with JWT session	A verified student account exists.	1. Submit valid credentials to POST /api/auth/login. 2. Inspect the response body and headers. 3. Decode the returned JWT. 4. Call a protected endpoint with the token. 5. Repeat with an invalid password.	Valid credentials return HTTP 200 with a signed JWT containing user id, role and expiry; the protected endpoint returns 200. Invalid credentials return 401 with a generic error and no token. (AC-003)	Functional / High
TC-Auth-05	IPP-F-005	Account lockout after 5 failed login attempts	A verified account exists and is not currently locked.	1. Attempt login with an incorrect password five consecutive times. 2. Attempt a sixth login with the correct password. 3. Wait 15 minutes and log in with the correct password.	Attempts 1–5 return "Invalid credentials". The 6th attempt is rejected with a lockout message even though the password is correct. After 15 minutes login succeeds. (AC-005)	Functional / High
TC-List-01	IPP-F-006	Verified recruiter posts an internship listing	A recruiter account approved by the admin is logged in.	1. Open "Post Internship". 2. Fill title, description, required skills, stipend, duration, location and deadline. 3. Submit. 4. Log in as a student and search for the listing title within 1 minute.	The listing is saved, confirmation is shown, and the listing appears in student search within 1 minute with all fields intact. (AC-006)	Functional / High
TC-List-02	IPP-F-007	Search and filter internships	At least 20 listings exist with varied skills, locations,	1. Log in as a student and open the internship search page.	Every returned listing satisfies all four filter criteria; non-matching	Functional / High

TC ID	Req ID	Objective	Preconditions	Test Steps	Expected Result	Type / Priority
			stipends and durations.	2. Apply filters: skill = "React", location = "Bengaluru", stipend $\geq$ 10000, duration = 3 months. 3. Inspect every returned record. 4. Apply a filter combination with no matches.	listings are excluded. The empty combination shows a "No internships found" message rather than an error. (AC-007)	
TC-App-01	IPP-F-010	Apply to an internship with resume upload	Student is logged in; an open listing exists; a valid PDF resume is available.	1. Open a listing and click Apply. 2. Upload a valid PDF resume and enter an optional cover letter. 3. Submit. 4. Verify the application record in the database and on the dashboard. 5. Attempt to apply to the same listing again.	An application record is created with status "Applied" and a timestamp; the resume is stored and retrievable. A duplicate application is rejected with an appropriate message. (AC-010)	Functional / High
TC-App-02	IPP-F-011	Track application status in real time	A student has an application with status "Applied"; the recruiter account for that listing is available.	1. Note the status on the student dashboard. 2. As the recruiter, change the status to "Shortlisted". 3. Refresh the student dashboard. 4. Repeat for "Selected".	The student dashboard reflects the recruiter's latest status after refresh, with the correct status label and update timestamp in every case. (AC-011)	Functional / High
TC-Test-01	IPP-F-014	Timed aptitude test auto-submits at zero	Student is logged in; a quantitative test of 10 questions with a 10-minute timer is configured.	1. Start the test and confirm the countdown timer is visible. 2. Answer 6 questions and leave 4 unanswered. 3. Allow the timer to run to 00:00 without submitting. 4. Observe the resulting screen and the stored attempt record.	The timer counts down continuously; at 00:00 the test auto-submits, no further input is accepted, and the attempt is recorded with the 6 answered questions. (AC-014)	Functional / High
TC-Test-02	IPP-F-015	Auto-grading and immediate score display	A test with a known answer key is available.	1. Take a 10-question test and deliberately answer exactly 7 correctly. 2. Submit the test. 3. Compare the displayed score with the answer key. 4. Check the attempt history entry.	The score 7/10 is displayed immediately after submission and matches the correct-answer count; the same score and the time taken appear in attempt history. (AC-015, AC-016)	Functional / High
TC-Admin-01	IPP-F-020	Admin approves and rejects recruiter	Two recruiter accounts are in the pending state; admin	1. Open the pending recruiter queue. 2. Approve recruiter A and reject	Recruiter A becomes verified and can post listings. Recruiter B	Functional / High

TC ID	Req ID	Objective	Preconditions	Test Steps	Expected Result	Type / Priority
		registrations	is logged in.	recruiter B with a reason. 3. Log in as recruiter A and attempt to post a listing. 4. Attempt to log in as recruiter B.	cannot log in or post listings and sees the rejection message. (AC-020, AC-002)	
TC-Perf-01	IPP-NF-001	Search and page-load response time under normal load	Database seeded with 1,000 listings; JMeter configured for the normal-load profile (200 concurrent users).	1. Run a 10-minute JMeter test exercising login, search with filters, listing details and dashboard load. 2. Record the 90th-percentile response time per endpoint. 3. Record the error rate.	The 90th-percentile response time is ≤ 3 seconds for every measured endpoint and the error rate is below 1%.	Performance / High
TC-Perf-02	IPP-NF-004	Concurrency at 5,000 virtual users	Load environment provisioned to production-equivalent capacity.	1. Ramp JMeter to 5,000 virtual users over 10 minutes. 2. Sustain the load for 20 minutes on the search and dashboard flows. 3. Monitor response times, CPU, memory and database connections.	The system remains available for 5,000 concurrent users without noticeable performance degradation, excessive errors, or resource exhaustion. Any deviation from full scale is documented.	Performance / Medium
TC-Sec-01	IPP-NF-003, IPP-SR-002	Password storage uses salted bcrypt	Direct read access to the test database is available.	1. Register a new user with a known password. 2. Query the users collection and inspect the password field. 3. Verify the hash prefix and that two users with identical passwords have different hashes. 4. Search application logs and API responses for the plaintext password.	Only a bcrypt hash (prefix \$2a\$/\$2b\$) is stored; identical passwords produce different hashes, proving per-user salting; the plaintext never appears in logs, responses or the database.	Security / High
TC-Sec-03	IPP-SR-003	RBAC enforced on every API endpoint	Valid student, recruiter and admin tokens are available; the endpoint inventory is documented.	1. Call each protected endpoint with no token. 2. Repeat with an expired token and a token whose signature has been tampered with. 3. Call admin endpoints with a student token and recruiter-scoped endpoints with another recruiter's token.	Every request without a valid token returns 401; every cross-role or cross-owner request returns 403. No endpoint leaks data to an unauthorised role.	Security / High
TC-Sec-04	IPP-SR-004	Aptitude test content served via time-bound	A student is enrolled in a test session.	1. Start a test and capture the content-request token.	Replayed and expired tokens are rejected; test content cannot be	Security / High

TC ID	Req ID	Objective	Preconditions	Test Steps	Expected Result	Type / Priority
		tokens		2. Replay the same token in a new session. 3. Use the token after its validity window has expired. 4. Inspect the client-side payload for answer keys.	fetched without a valid live token; no correct answers are present in the client payload before submission.	
TC-UX-01	IPP-NF-005	Responsive layout and WCAG 2.1 AA basics	Build deployed; axe DevTools installed; screen reader available.	1. Load the student dashboard, search page and test page at 1920px, 768px and 375px widths. 2. Run an axe automated scan on each page. 3. Navigate the apply flow using only the keyboard. 4. Check colour contrast and form-label associations.	Layouts reflow without horizontal scrolling or overlap at all three widths; no critical or serious axe violations; the full apply flow is keyboard-operable with a visible focus indicator; contrast meets 4.5:1.	Usability / Medium

Total: 16 detailed test cases — 10 functional (covering registration, login, logout, listing creation, search and filtering, application submission, status tracking, timed testing, auto-grading and admin recruiter approval) and 6 non-functional (2 performance, 3 security, 1 usability/accessibility). Remaining test case IDs referenced in Section 3 and the RTM are specified in the Jira test repository following the same format.